

Bioethics: Science and the *Future of Medicine*

A discussion-led (Harkness) course on the ethics of modern biology, genetics, and medicine. Students prepare through readings and videos, debate the questions in class, and complete a final project — a research paper or a presentation — shared at a closing panel.

10

ONE-HOUR SESSIONS

10

STUDENTS PER CLASS

13+

RECOMMENDED AGE

Harkness

DISCUSSION FORMAT

UNIT 1 Foundations of Biology & Genetics

1 Cells, DNA & Genes

The essential biology behind genetic technologies: what genes are and how DNA carries hereditary information.

2 Inheritance & Designing Experiments

How traits are inherited through genes, mutation, and variation — and what makes scientific evidence trustworthy.

3 Harkness Discussions

Two debates: “*Should we test embryos for disease?*” and “*Should parents be able to choose their child’s traits?*”

UNIT 2 The History of Bioethics

4 From Nuremberg to Modern Ethics

How post-WWII abuses created modern research ethics, and how ideas about human difference were once used to justify discrimination.

UNIT 3 Bioethics in the Modern Era

5 CRISPR & Gene Therapy

How gene editing works, and where the line falls between curing disease and reshaping human beings.

6 AI in Healthcare

How AI is used in medicine today, and the concerns it raises around bias, privacy, and accountability.

UNIT 4 Project-Based Learning & Panel

7 Project Introduction & Proposal

Choosing a bioethics topic (e.g. 3D bioprinting) and a format — a research / review paper, or a presentation (a lighter option, recommended for less experienced students) — then drafting a proposal.

8 Work with Instructor I — *optional*

Guided work on the chosen project: developing the argument and strengthening the evidence.

9 Work with Instructor II — *optional*

Revising and finalizing the paper or presentation, with optional competition pathways (e.g. Scholastic essays).

10 Panel Presentation

Students present and defend their work before a panel and audience. Parents welcome to attend.